

PAYAL CHAUDHARY

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EDUCATION

PhD Candidate in Chemical Engineering

University of Nebraska-Lincoln (UNL), Department of Chemical and Biomolecular Engineering

2021 – Present
Lincoln, NE, USA

Integrated Dual Degree (B.Tech. + M.Tech.) in Materials Science and Engineering

Indian Institute of Technology (BHU) Varanasi, School of Materials Science and Technology

2014 – 2019
Varanasi, India

RESEARCH EXPERIENCE

Research Assistant

Department of Chemical and Biomolecular Engineering, University of Nebraska-Lincoln (UNL)

2021 – Present
Lincoln, NE, USA

- Advisor: Vitaly Alexandrov, PhD
- Research Interests: Computational study of oxygen evolution reaction (OER), water oxidation reaction (WOR), nitrate reduction reaction (NO_3RR), and the role of EDL on catalyst materials such as metals, metal oxides and graphene-based single- and dual-atom PGM-free catalysts. Integration of machine-learning methods to investigate transient dissolution of metal oxides at electrochemical conditions.

M.Tech. Dissertation

School of Materials Science and Technology, Indian Institute of Technology (BHU) Varanasi

2018 – 2019
Varanasi, India

- Advisor: Sanjay Singh, PhD
- Topic: Search for potential Weyl semimetal candidates in Heusler alloys using first-principle methods.

B.Tech. Project

Department of Physics, Indian Institute of Technology (BHU) Varanasi

2016 – 2017
Varanasi, India

- Advisor: Shradha Mishra, PhD
- Topic: Dynamics of a particle moving on Lorentz Lattice Gas on a random 2D collection of points.

INTERSHIPS

Computational Chemistry Materials Science (CCMS) - Graduate Intern

Materials Science Division, Lawrence Livermore National Laboratory (LLNL)

2026
Livermore, CA, USA

- Mentor: Timofey Frolov, PhD
- Topic: Grain boundary phase engineering and alloy survivability in fusion energy environment.

Visiting Student - Graduate

Materials Science Division, Argonne National Laboratory (ANL)

2025
Lemont, IL, USA

- Mentors: Peter Zapol, PhD and Pietro Papa Lopes, PhD
- Topic: Investigating atomistic dissolution mechanisms of electrochemical interfaces.

Computational Chemistry Materials Science (CCMS) - Graduate Intern

Materials Science Division, Lawrence Livermore National Laboratory (LLNL)

2024
Livermore, CA, USA

- Mentor: Sneha Akhade, PhD
- Topic: Developing descriptors for electrochemical conversion of nitrogen to ammonia.

Summer Research Project

Department of Materials Engineering, Indian Institute of Science (IISc)

2017
Bangalore, India

- Advisor: Abhik Choudhary, PhD
- Topic: Computing the phase diagrams of binary alloys using the BINGSS source code (Fortran implementation of the CALPHAD method).

Summer Research Project

Mechanical Metallurgy Division, Bhabha Atomic Research Centre (BARC)

2016
Mumbai, India

- Advisor: Dinesh Srivastava, PhD
- Topic: Investigating the effect of precipitate morphology and distribution on the mechanical behavior of binary Ni-Cr alloy.

Google Summer of Code

Participant with Syssters, an Anita Borg Institute Community

2016
India

- Improved the meetup application on Syssters' open-source web portal using Django, a python-based web framework. Link to the project: [archive](#).

HONORS AND AWARDS

CHME Best Paper Award: Second place University of Nebraska-Lincoln (UNL)	2026
Center for the Integration of Research, Teaching and Learning (CIRTL) Associate Certification UNL	2025
UNL Research Days – Engineering Poster Competition: Second Honors University of Nebraska-Lincoln (UNL)	2025
Milton Mohr Fellowship University of Nebraska-Lincoln (UNL)	2025
Othmer Fellowship University of Nebraska-Lincoln (UNL)	2021 – 2024
IASc-INSANA Summer Research Fellowship Program (SRFP) India	2016

ACADEMIC CONFERENCES

- AIChE Annual Meeting 2025, November 2-6, 2025, Boston, MA (oral, poster).
- 248th ECS Meeting, October 12-16, 2025, Chicago, IL (oral).
- AIChE Annual Meeting 2024, October 27-31, 2024, San Diego, CA (poster).
- ECS PRIME 2024, October 6-11, 2024, Honolulu, HI (oral).
- AIChE Annual Meeting 2023, November 5-10, 2023, Orlando, FL (oral).
- ASEE Midwest Section Conference 2023, September 10-12, 2023, Lincoln, NE (student panelist).
- ACS Fall 2023, August 13-17, 2023, San Francisco, CA (oral).

WORKSHOPS AND SUMMER SCHOOLS

Telluride School on Theoretical Chemistry Telluride, CO	2025
Principles of Learning Workshop, UNL Lincoln, NE	2025
AIChE Future Faculty Mentoring Program Mentors: Profs. Matteo Cargnello, Elizabeth Corson	2024
UNL Preparing Future Faculty (PFF) Program Mentor: Prof. Michael Janik	2024
Seminar on Grant Proposal Writing: Getting Started Lincoln, NE	2024
Career Discovery: How to be a Great ChE Professor Workshop, AIChE Annual Meeting San Diego, CA	2024
Effective Teaching for New or Prospective Faculty Workshop, AIChE Annual Meeting Orlando, FL; San Diego, CA	2023-24
The 5th i-CoMSE Workshop: Machine Learning for Molecular Science Minneapolis, MN	2023
Summer School for Integrated Computational Materials Education Ann Arbor, MI	2023

TEACHING AND MENTORING EXPERIENCE

Teaching Assistant	2024, 2026
Department of Chemical and Biomolecular Engineering, University of Nebraska-Lincoln	Lincoln, NE, USA
– Chemical Engineering Computation	
– Transport Operations III	
Mentoring High School Intern	2024 – 2025
Department of Chemical and Biomolecular Engineering, University of Nebraska-Lincoln	Lincoln, NE, USA
Mentoring High School Summer Intern	2023, 2024, 2025
Department of Chemical and Biomolecular Engineering, University of Nebraska-Lincoln	Lincoln, NE, USA
– NCMN STEM High School Internship Program	
Teaching Assistant	2018 – 2019
School of Materials Science and Technology, Indian Institute of Technology (BHU) Varanasi	Varanasi, India
– Physical Behavior of Materials	
– Introduction to Engineering Materials	

PUBLICATIONS

Google Scholar: [Payal Chaudhary](#)

Journals

Denotes equal contributions

1. R. Jana, O. Q. Carvalho, R. Bello, **P. Chaudhary**, J. Hong, A. S. Hoffman, S. R. Bare, A. Gupta, V. Alexandrov, K. A. Stoerzinger, "Disentangling the impact of cation identity on transport and kinetics in nitrate electroreduction to ammonia", *ChemRxiv* 2026.
2. D.A. Kuznetsov, V. Vogt, P.M. Abdala, D.D. Khalyavin, C.W. Galdino, **P. Chaudhary**, V. Alexandrov, Y. Lu, L. Jin, F. Donat, P. Manuel, W. Zhang, Z. Zhang, T. Schmitt, S. Tippireddy, C. R. Müller, "Intramolecular oxygen coupling mediates water oxidation on an oxide electrocatalyst", *ChemRxiv* 2026.

3. Y. Xu, Y. H. Wu, P. M. Abdala, C. Sherwin, V. Celorrio, D. Piankova, **P. Chaudhary**, V. Alexandrov, A. M. Kierzkowska, D. A. Kuznetsov, C. R. Müller, "Effect of Metal Substitution on the Intrinsic Activity of Iridium-Based Oxides for the Oxygen Evolution Reaction", *Chemical Science* 2025, 16, 47, 22757-22768.
4. Z. Lin, **P. Chaudhary**, S. A. Vigil, M. Fratarcangeli, C. J. Soderstedt, V. Alexandrov, I. A. Moreno-Hernandez, "Understanding the reaction energetics of oxygen-evolving electrocatalysts", *Energy & Environmental Science* 2025, 18, 16, 8029-8038.
5. **P. Chaudhary**, V. Alexandrov, "Mechanism of Bicarbonate Enhanced Solar Oxidation of Water to Hydrogen Peroxide over BiVO₄", *ACS Catalysis* 2025, 15, 9006-9012.
6. S. Xue[#], **P. Chaudhary**[#], M. R. Nouri, E. Gubanovaa, B. Garlyyev, V. Alexandrov, A. S. Bandarenka, "Impact of Pt(*hkl*) Electrode Surface Structure on the Electrical Double Layer Capacitance", *Journal of the American Chemical Society* 2024, 146, 6, 3883-3889.
7. **P. Chaudhary**, A. Zagalskaya, H. Over, V. Alexandrov, "Strain–Dependent Activity–Stability Relations in RuO₂ and IrO₂ Oxygen Evolution Catalysts", *ChemElectroChem* 2023, e202300659.
8. A. Zagalskaya[#], **P. Chaudhary**[#], V. Alexandrov, "Corrosion of Electrochemical Energy Materials: Stability Analyses Beyond Pourbaix Diagrams", *Journal of Physical Chemistry C* 2023, 127, 30, 14587-14598 – Invited Perspective; [Cover Page](#).
9. **P. Chaudhary**, I. Evazzade, R. Belosludov, V. Alexandrov, "Computational Discovery of Active and Selective Metal-Nitrogen-Graphene Catalysts for Electrooxidation of Water to H₂O₂", *ChemCatChem* 2023, 15, e202300055.
10. **P. Chaudhary**, K.K. Dubey, G.K. Shukla, S. Singh, S. Sadhukhan, S. Kanungo, A.K. Jena, S.C. Lee, S. Bhattacharjee, J. Minár, S.W. D'Souza, "Role of chemical disorder in tuning the Weyl points in vanadium doped Co₂TiSn", *Physical Review Materials* 2021, 5, 12, 124201.
11. M. Singh, S. Kumar, M. Alam, V.K. Gangwar, L. Ghosh, D. Pal, R. Singh, P. Shahi, **P. Chaudhary**, K. Shimada, S. Chatterjee, "Evidence of surface and bulk magnetic ordering in Fe and Mn doped Bi₂(SeS)₃ topological insulator", *Applied Physics Letters* 2021, 118, 13.

SERVICE AND OUTREACH

Adsorption and Reactions on Catalyst Surfaces: Exp. and Theor. Approaches, Session Co-Chair	2026
2026 AIChE Annual Meeting	Minneapolis, MN
Photocatalysis II: Applied Photocatalysis for Environmental & Chemical Conversions, Session Co-Chair	2025
2025 AIChE Annual Meeting	Boston, MA
Student Research Days Undergraduate Poster Session and Creative Exhibition, Judge	2023
University of Nebraska-Lincoln	Lincoln, NE
Undergraduate Creative Activities and Research Experience (UCARE) Program, Applications Reviewer	2023
University of Nebraska-Lincoln	Lincoln, NE
Student Alumni Mentorship Program, Mentor	2023
Indian Institute of Technology (BHU) Varanasi	Varanasi, India
Conference for Undergraduate Women in Physical Sciences (WoPhys), Volunteer	2021
University of Nebraska-Lincoln	Lincoln, NE
Department Undergraduate Committee (DUGC), Member	2017 – 2018
School of Materials Science and Technology, Indian Institute of Technology (BHU) Varanasi	Varanasi, India